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ORIGINAL ARTICLE

Using risk factors, myoelectric signal, and finger tremor to distinguish computer users with and without musculoskeletal symptoms

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Abstract Most of previous studies use questionnaire to assess risk factors for cumulative trauma disorders of the upper extremity (CTDUE) for computer workers. Few studies combine both physical examination and questionnaire to assess musculoskeletal symptoms. Fifteen symptomatic and 15 non-symptomatic computer users were recruited. Both of them were asked to perform a repetitive tapping task (200 taps/min) as the fatigue task. Tremor of the index finger and surface electromyography (EMG) of the flexor digitorum superficial (FDS) were collected prior and after the tapping task. Muscle strength and range of motion for right wrist were collected before the tapping task. All subjects were asked to fill out the questionnaire about risk factors of CTDUE. Female users in the symptomatic group had weaker wrist extensor strength than those in the non-symptomatic group ($P < 0.05$). After performing the tapping task, FDS strength and median frequency of the FDS EMG at 25%, and 100% maximal voluntary contraction (MVC) for the symptomatic group decreased ($P < 0.05$). However, no significant differences were found in strength and median frequency of the FDS EMG between prior and after tapping task in the non-symptomatic group. There was no significant difference for root mean square of the finger tremor between and within groups. Regression analysis revealed that median frequency of the FDS EMG at 25% MVC, age, total time spent on computer, and mouse position were better factors to classify computer users into the symptomatic group compared to other factors obtained from questionnaire and physical exam. Besides, symptomatic computer users seem to have longer experience of computer use than non-symptomatic users. After the fatigue task, the symptomatic users decreased their muscle strength to a larger extent than the non-symptomatic users.

Keywords Musculoskeletal symptoms · Risk factors · Cumulative trauma disorders · Fatigue · Median frequency

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Introduction

Cumulative trauma disorder of upper extremity (CTDUE) is an umbrella term used to describe disorders resulting from repetitive use of the upper limb over time rather than a specific incident (Stobbe 1996). The advent of personal computer in our daily life has drawn our attention on work related disorder of the upper extremity might be linked to computer use. Prevalence of CTDUE in computer users is about 15% (Pascarelli and Hsu 2001). CTD in computer users includes tendonitis, carpal tunnel syndrome, forearm pain, finger tremor, and dysfunction of sympathetic nervous system. Since CTDUE can become irreversible, this problem cannot be neglected (Pascarelli and Hsu 2001).

Different risk factors might be related to CTDUE which includes personal factors, job factor, and environmental factors. Potential risk factors specific to job include repetitious work, awkward or sustained postures, cold temperature, time pressure, monotonous work, and vibration. Personal factors which might be related to CTDUE include old age, female gender, smaller carpal canal size, elevated body mass index, and previous musculoskeletal consultation (Leclerc et al. 2001; Zakaria et al. 2002). For the computer users, previous studies (Punnett and Bergqvist 1999) suggested that intensive keyboard and mouse use as well as postural stress are associated with upper extremity muscu-

loskeletal disorder. Kryger et al. (2003) suggested that individuals who spent more than 15 h/week on keyboarding or people who spent more than 10 h/week on mouse use tend to have a higher rate of CTDUE. Low co-worker support as well as history of musculoskeletal disorders were also related to high incidence of CTDUE (Jensen 2003).

For the musculoskeletal symptoms induced by prolonged use of computer, it is not easy to have a direct diagnosis from history taking or questionnaire, more detailed examinations are required. Early detection of risk factors is also important to prevent CTDUE. Previous studies have applied sensory threshold test, muscle strength assessment, range of motion test, as well as electro-physiologic procedure to assist in differentiation people with musculoskeletal symptoms. Jensen et al. (2002) suggested that sensory threshold of vibration on the right hand was higher for the computer users than the age matched control. Subjects with musculoskeletal symptoms even had a higher sensory threshold in the radial nerve innervation area. The other study suggested that people without symptoms had a higher sensory threshold in the median and ulnar nerve innervation area (Pilegaard and Jensen 2005). Female computer users with musculoskeletal symptoms had a lower horizontal pinch-grip value than non-symptomatic group. As for range of motion test, no significant difference was found between two groups (Jensen et al. 2002). These studies suggested that some of the physical examinations such range of motion assessment might not be able to differentiate people who were at risk of CTD, physiologic examination procedure such as electromyography (EMG) and tremor assessment might be better indicators to explore the nature of CTDUE.

Fatigue has a major impact on motor function, altering both central and peripheral components of the control system. It is common to detect muscle fatigue among prolonged computer users, especially the extensor digital groups (Lin et al. 2004). The EMG frequency shift has been successfully used to examine muscle fatigue (Cram et al. 1998). During sustained muscle contractions, the decrease in EMG median frequency is related to the decrease in conduction velocity of motor potentials on the sarcolemma (Basmmajian and Deluca 1985). Arihara and Sakamoto asked the subjects to maintain middle finger in a stretch position with 200 g loading for 10 min as a fatigue task. They found the shape of the EMG spectrum of the extensor digitorum communis gradually shift to a lower domain after fatigue. In addition to sustained muscle contraction, repetitive movement could also elucidate fatigue. Morrison et al. (2005) asked the subjects to continuously flex and extend their wrist against moderate resistance, and they found the majority of the power in the EMG signal of extensor digitorum muscle was shifted to a lower frequency domain, between 50 and 150 Hz. Takanokura et al. (2001) asked their subjects to continuously tap on a button for at least 30 min. They found the power spectrum of the flexor digitorum reaches to its maximal amplitude after 30 min tapping; however, it decreased after 60 min. In addition to EMG spectrum analysis, previous studies also investigate finger tremor after fatigue. They found that the power spectrum of finger tremor would shift to that range of 8–12 Hz after performing a fatigue task (Arihara and Sakamoto 1999; Takanokura et al. 2001). According to previous literature, EMG and tremor analysis might be good indicators for muscular fatigue.

Some studies have examined the effect of typing on finger flexors and extensors by using EMG after a long period of typing task (Lin et al. 2004; Szeto et al. 2005). Lin et al. found the median frequency of extensor digitorum decreased by 25% in comparison with the initial value after a 2-h typing task. Szeto et al. found that symptomatic computer users consistently showed trends for higher median frequency of the neck and shoulder muscles than controls as they increased their typing speed. The cumulative effect of mouse use is also a risk factor for CTDUE but it was seldom investigated. Most of the previous literature use questionnaire to investigate risk factors of CTDUE (Jensen 2003; Kryger et al. 2003). We speculate that the symptomatic subject might alter their motor control pattern after a fatiguing task that might require investigation of the underlying mechanism. The purpose of this study was to compare the differences between the symptomatic and non-symptomatic computer users in the risk factors as well as clinical signs and symptoms of CTDUE. Specifically, we would like to find out which factor(s) might be better to distinguish people with and without musculoskeletal symptoms. We hypothesized that after performing a continuous tapping task, people with musculoskeletal symptoms were easier to show muscle fatigue as well as finger tremor than those without musculoskeletal symptoms. We also hypothesized that some of these risk factors as well as clinical signs and symptoms may be useful factors to distinguish computer users who have the complaint of musculoskeletal symptoms from users without symptoms.

Method

Subjects

Fifteen symptomatic and 15 non-symptomatic right-handed computer workers volunteered for this study. All of them exposed to computer work more than 3 h/day. The symptomatic group had to report musculoskeletal symptoms primarily in their right forearm or hands at least 8 days within the past year. Moreover, they had to report at least 2/10 on a numerical scale of 0–10 (0 = no symptom, 10 = cannot

tolerate) within the last 3 months. The non-symptomatic group had no symptom in their right forearm or hands.

Questionnaire

Subjects were asked to complete a questionnaire. The questionnaire contains four parts: basic information, preferred ergonomic placement of workstation, self-described musculoskeletal symptoms, and mental health. The basic information of questionnaire included personal characteristics such as age, sex, body mass index (BMI), recreation time, experiences of computer use. Basic information of questionnaire was designed to investigate what kinds of static and dynamic recreational activities they participated. Also we would like to know how much time they spent on computer work (Cho et al. 2003; Burke and Peter 2002). The preferred ergonomic placement of workstation was designed to obtain information about preference of workstation arrangement, including: table height, chair height, positions of keyboard and mouse; whether their chair or table could be adjustable (yes or no), and whether mouse and keyboard were placed on the same level (yes or no) were also recorded. Self-described musculoskeletal symptoms were obtained by using musculoskeletal symptom questionnaire (Cho et al. 2003) which was modified from Nordic questionnaire (Kurrinka et al. 1987). Mental health was designed investigate psychological factors and used 12-item versions of the Chinese-Health-Questionnaire (Chong and Wilkinson 1989). According to the data from previous studies, work station arrangement was classified as "normal" or "abnormal" (Ketola et al. 2004; Kroemer and Grandjean 1997; Kryger et al. 2003). Normal table and chair height were suggested between 65 and 82 cm and 43 and 57 cm respectively. Abnormal keyboard position was defined as a keyboard was not placed in front of the trunk. Abnormal mouse position was defined as a mouse was placed more than 40 cm to the edge of the table. Also, if the score of Chinese-Health-Questionnaire was higher than five, the person was classified into a high mental distress group.

Muscle strength and range of motion

Muscle strength of maximal voluntary contraction (MVC) of right wrist flexor, wrist extensor and flexor digitorum superficial (FDS) were measured with a strain-gauge transducer (sensitivity 0.1 N, Model 9800, Aikoh, Japan). To measure strength of wrist flexors and extensors, subjects placed their forearm in supination or pronation, respectively. Then, subjects were asked to perform wrist flexion or extension movement to resist strain-gauge transducer in the distal end of metacarpal bone for 5 s. To measure strength of flexor digitorum superficial, subjects placed their forearm in neutral position with index finger proximal inter-phalangal (PIP) joint flex and distal-phalangal (DIP) joint extend pressing the strain-gauge transducer for 5 s. Three MVC recordings were required and 30 s rest between measurements.

Range of motion of wrist flexion/extension, and ulna/radial deviation were measured with goniometer in standard position.

Finger tremor and EMG measurement

Finger tremor of index finger was measured with two Qualisys motion capture system cameras (Qualisys AB, Gothenburg, Sweden) before and after tapping tasks. Subjects sat on a chair with the elbow extended and forearm pronated on the table. They were asked to conduct the pointing task with index finger extension and other fingers flexion/adduction to target a black point for 10 s in front of them. Each measurement was repeated for three times. A marker was placed on the dorsal side of index finger distal phalange. The cameras were sampled at 240 Hz. Raw data was then computed using Qualisys tracking manager software with 25 points moving average filtering and calculated acceleration in three-dimensional movement. The acceleration data was exported to EXCEL and we calculated root mean square (RMS) to represent magnitude of index finger tremor.

Muscle activity of FDS was recorded by a pre-amplified bipolar surface electrode (Motion Control, Inc., Salt Lake City, USA) (electrode spacing 2.5 cm, diameter 1.1 cm, with a gain of 375 and a common-mode rejection ratio (CMRR) of 102 dB) (Motion Control, Inc., UT, USA) that were placed on the ventral side of forearm. The electrode was positioned approximately three quarters of the distance from elbow to wrist (Cram et al. 1998). Subjects sat comfortably on a chair and placed forearm in neutral position with index finger PIP joint flexion and DIP joint extension. They were tested by EMG while pressing the strain-gauge transducer in 25% MVC for 30 s, 60% MVC, and 100% MVC for 10 s respectively, both before and after tapping task. Subjects had to watch the screen of strain-gauge transducer as a feedback and were required to do their best to maintain the same amount of force continuously. After each trial, the subjects had 30 s break. The sequence of the test was randomly selected. EMG signals were sampled at 1,000 Hz and filtered with an eighth order Butterworth band pass filter (2–400 Hz). Median frequency and RMS of the FDS EMG were computed by Matlab software (The Math Works, Inc., USA) as the outcome measures. The RMS approached the quantification of EMG signal by squaring the data, summing the squares, dividing this sum by the number of observations, and finally taking the square root (Cram et al. 1998).

Experimental procedure of tapping task

After muscle strength and range of motion tests, subjects rested for 5 min and performed a fatigue task. For the fatiguing task, subjects were asked to conduct a tapping task on a computer mouse (200 taps/min) in different period of time according to his own MVC (Acer, MS.MUV01.003). It was approximately 2.2 N for one keystroke. We computed the tapping time individually from the equation of 15% MVC of FDS × 200 taps × 5 min = Resistance of keystroke (2.2 N) × 200 taps × Tapping time. For example, if a person's MVC is 30 N, 30 × 15% × 200 × 5 (min) = 2.2 × 200 × tapping time. Tapping time = 10 min (Morrison et al. 2005; Takanokura et al. 2001). Subjects sat in front of the computer with their right wrist rested on the edge of table without forearm support. Numbers of tapping and time were shown on the screen in order to control the velocity of tapping task. Finger tremor and EMG were measured before and after tapping task.

Statistical analysis

Independent *T*-test was used to compare the basic information of questionnaire, range of motion, median frequency of EMG and RMS of EMG and finger tremor between groups. Pearson correlation coefficients were calculated to determine the association between mental distress and musculoskeletal symptoms. In addition, paired *T*-test was used to compare EMG, finger tremor parameters and FDS muscle strength between before and after the tapping task. Non-parametric test was used to compare muscle strength between male and female groups respectively. Chi-square test was used to compare preferred ergonomic placement of workstation and mental health of questionnaire. The "abnormal position" and "high mental distress" were scored as "1," while "normal position" and "low mental distress" were scored as "0." In order to find out important factors to distinguish computer workers with musculoskeletal symptom from people without symptoms, linear regression method was used. Significant level was set at 0.05 for all tests. All of the data were analyzed by SPSS 13.0 for windows (Chicago, IL, USA).

Results

Questionnaire

For the demographic data (Table 1), there were no significant differences in age, height, weight, BMI, time spent on computer per week and recreational time between two groups. However, symptomatic group had longer experience for computer use than non-symptomatic group ($P = 0.04$).

Table 1 Basic information of questionnaire

	Symptom	Non-symptom
Age (years)	23.33 ± 1.29	24.00 ± 1.73
Height (m)	1.69 ± 0.09	1.70 ± 0.07
Weight (kg)	66.50 ± 12.44	61.03 ± 9.77
BMI (kg/m^2)	23.16 ± 3.27	21.12 ± 2.70
Recreation time (h)	12.14 ± 5.06	14.61 ± 9.54
Time spent (h/week)	50.03 ± 22.50	41.07 ± 16.45
Experience of computer use (years)*	7.63 ± 2.33	6.13 ± 1.42

*$P < 0.05$

For the preferred ergonomic placement of workstation (Table 2), there were no significant differences between groups ($P > 0.05$). However, the number of people in the symptomatic group who used computer in abnormal position was larger than that in the non-symptomatic group. For the self-described musculoskeletal symptoms (Table 3), symptomatic group reported the worst symptom occurred to the wrist region, following by finger, forearm and elbow. Finally, for the mental health assessment, three subjects in the symptomatic group and none of the non-symptomatic were classified into high mental distress group (CHQ ≥ 5). The association between mental distress and severity of musculoskeletal symptom was high ($r = 0.601$, $P < 0.0001$).

Muscle strength and range of motion

It was found that muscle strength of wrist flexors and FDS were not significant different between symptomatic and non-symptomatic groups. Further analysis revealed that females in the symptomatic group had weaker strength of wrist extensors than the non-symptomatic group ($P = 0.047$), but there was no significant difference in for two male groups (Table 4).

Table 2 Preferred ergonomic placement of workstation

Compared with normal range	Symptom (n = 15)		Non-symptom (n = 15)	
	Normal	Abnormal	Normal	Abnormal
Table height	12	3	14	1
Chair height	5	10	8	7
Keyboard position	12	3	13	2
Mouse position (relative to edge of table)	13	2	15	0
Answer to following questions	Yes	No	Yes	No
Adjust table/chair	5	10	4	11
Mouse position (same level with keyboard)	10	5	11	4

Table 3 Self-described musculoskeletal symptoms for different body regions

	n	Symptom (*n* = 15) Points (range)	*n*	Non-symptom (*n* = 15) Points (range)
Elbow	7	2.07 (0–7)	0	0 (0–0)
Forearm	10	2.23 (0–6)	0	0 (0–0)
Wrist	10	3.23 (0–7.5)	0	0 (0–0)
Finger	8	2.37 (0–8)	0	0 (0–0)

After performing a tapping task, it was found that both groups decreased the strength of FDS after tapping task (Fig. 1). People in the symptomatic group significantly decreased the strength of FDS after the tapping task ($P = 0.004$), but the trend was not found in the non-symptomatic group ($P > 0.05$). There was also no significant difference for the measurement of range of motion between groups (Table 5).

EMG and finger tremor measurement

Compared to the non-symptomatic group, the symptomatic subjects significantly decreased the median frequency of the FDS EMG for 25% MVC after tapping tasks ($P < 0.001$), but they did not decrease the median frequency of the FDS EMG during 60 and 100% MVC (Fig. 2a). People in the symptomatic group significantly decreased their median frequency of the FDS EMG of 25% ($P < 0.001$) and 100% MVC ($P < 0.008$) (Fig. 2b), however, no significant difference was found in the non-symptomatic group (Fig. 2c). For the RMS analysis of EMG, the differences of 25% MVC between groups almost reached the significant level ($P = 0.079$) (Fig. 3a). There were no significant differences existed between two groups for other levels of contraction (Fig. 3b–d). For the finger tremor measurement, no significant differences were found in RMS of acceleration between and within groups (Fig. 4).

Useful factors to distinguish computer users with musculoskeletal symptoms

Linear regression from the results of the physical examination (Table 6) revealed that median frequency of the FDS EMG for 25% MVC was the only significant factor to distinguish computer users with musculoskeletal symptoms compared to other physical exam measures such as RMS and muscle strength. Linear regression from results of the questionnaire (Table 7) suggested that age, time spent on computer per week, experience of computer use as well as mouse position (relative to edge of table > 40 cm) were significant factors to distinguish computer users with musculoskeletal symptoms.

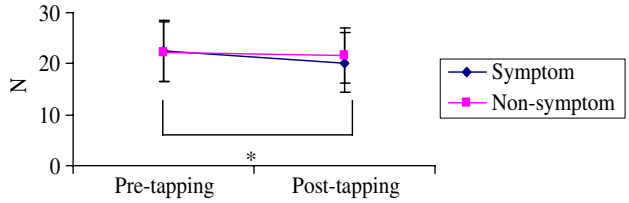

Fig. 1 Muscle strength of FDS before and after tapping task. (*$P < 0.01$ within the symptomatic group)

Table 5 Range of motion measurement

	Symptom	Non-symptom
Wrist flexion (°)	84.0 ± 4.8	82.0 ± 7.1
Wrist extension (°)	78.6 ± 6.8	80.9 ± 6.3
Ulnar deviation (°)	43.3 ± 13.0	43.5 ± 8.6
Radial deviation (°)	27.5 ± 8.7	25.7 ± 7.0

Discussion

The aim of this study was to compare the differences between symptomatic and non-symptomatic computer users in the risk factors as well as clinical signs and symptoms of CTDUE. Specifically, we would like to investigate which factor(s) could distinguish computer users with musculoskeletal symptoms. In this study, we used objective physical examinations to investigate the difference between groups. We found following a fatigue tapping task, a significant decrease in median EMG frequency of the FDS muscle for the symptomatic group, especially for the 25% MVC. Previous studies have similar finding that the median EMG frequency of the FDS muscle decreased after performing a fatigue task (Alizadehkhaiyat et al. 2007; Vigouroux and Quaine 2006). However, their subjects had to

Table 4 Muscle strength measurement

	Symptom		Non-symptom	
	Male (*n* = 10)	Female (*n* = 5)	Male (*n* = 10)	Female (*n* = 5)
FDS (N)	22.52 ± 3.64	17.73. ± 3.78	23.70 ± 4.31	17.54 ± 4.23
Wrist extensor (N)	61.43 ± 13.90	40.39 ± 15.61*	61.95 ± 11.73	61.41 ± 12.13*
Wrist flexor (N)	59.86 ± 11.01	47.35 ± 11.32	58.34 ± 13.61	58.62 ± 19.83

*$P < 0.05$

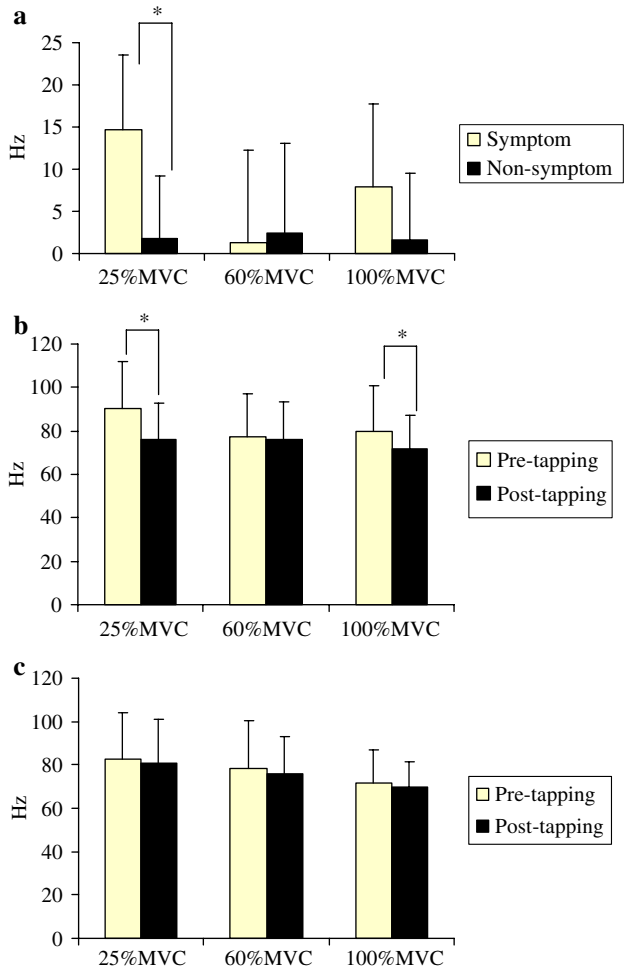

Fig. 2 **a** Changes of median frequency for different contraction levels. (*$P < 0.001$ between groups). **b** Median frequency of symptomatic group during different levels of contraction. (*$P < 0.008$). **c** Median frequency of non-symptomatic group during different levels of contraction

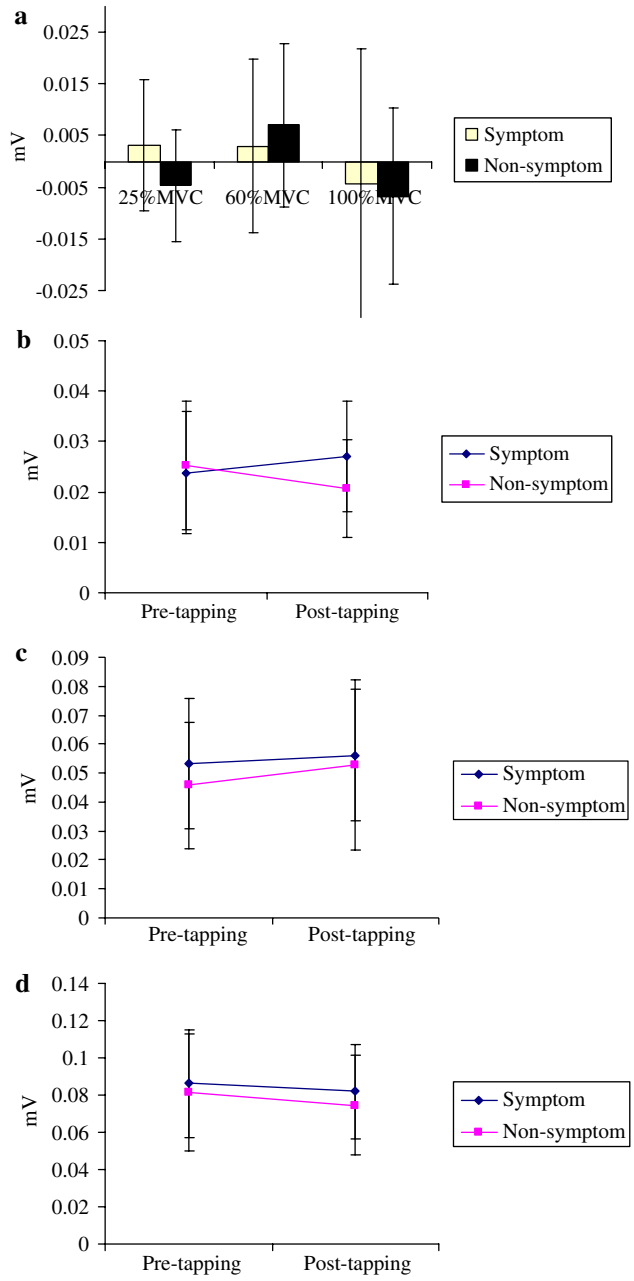

Fig. 3 **a** Root mean square of EMG changes (post RMS–pre RMS) during different contraction levels between groups. **b** Root mean square of EMG during 25% MVC before and after tapping task between groups. **c** Root mean square of EMG during 60% MVC before and after tapping task between groups. **d** Root mean square of EMG during 100% MVC before and after tapping task between groups

perform the finger flexion movement intermittently using higher level of effort (80 and 50% MVC). The participants in the current study had to perform a low intensity, repetitive tapping task using mouse and the duration to fatigue point was computed according to individual's MVC. A greater decrease in the median EMG frequency of the FDS muscle of the symptomatic group than the non-symptomatic group may suggest a greater rate of fatigue. It was also found that the median EMG frequency of the FDS muscle of the 25 and 100% MVC shifted to lower range only for the symptomatic group. We suspected that it was a sign of fatigue for the symptomatic group since this group might be easier to reach to their fatigue point. Interestingly, minimum change of the median EMG frequency of the FDS muscle was found for 60% MVC which was probably due to high variability of EMG within group.

Due to the high variability of the root mean square, no significant difference of RMS was found between groups. Subjects with symptoms slightly increased the RMS of 25% MVC after the tapping task; however, subjects without symptoms decreased their RMS after the task. Both groups might use different strategy to cope with this low intensity fatigue task. On the contrast, both groups had the same tendency to change their RMS for higher level of MVC. High intensity of muscle contraction might not be easily maintained after the fatigue task.

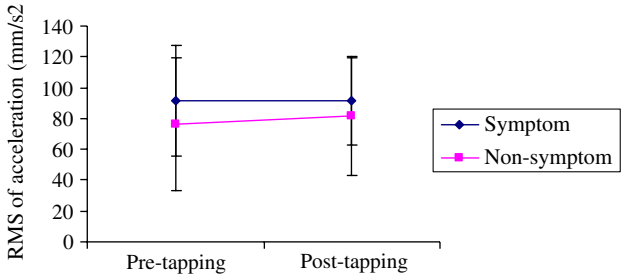

Fig. 4 Root mean square of acceleration before and after tapping task between groups

Table 6 Linear regression analysis from results of the physical examination (MF 25: changes of median frequency for 25% MVC after performing the tapping task. RMS 25: changes of root mean square for 25% MVC after performing the tapping task)

Independent predictors	β	P
MF 25	0.029	0.001*
MF 60	−0.007	0.373
MF 100	0.004	0.694
RMS 25	−10.705	0.155
RMS 60	1.927	0.696
RMS100	−1.527	0.697
Constant	0.283	0.013

*$P < 0.05$

Table 7 Linear regression analysis from results of the questionnaire

Independent predictors	β	P
Age	−0.155	0.006*
Recreation time	−0.014	0.119
Time spent on computer per week	0.018	0.001*
Experience of computer uses	0.176	0.001*
BMI	−0.001	0.963
Table height	0.316	0.200
Chair height	0.214	0.119
Adjust table/chair	0.161	0.380
Keyboard position	0.269	0.155
Mouse position (same level with keyboard)	−0.171	0.290
Mouse position (relative to edge of table)	1.452	0.001*
Mental health	−0.259	0.327
Constant	2.078	0.147

*$P < 0.05$

We did not find significant increase of finger tremor after the tapping task which was different from previous study (Huang et al. 2006). Previous study suggested that RMS of finger acceleration would increase as the muscle contraction time lasted longer. Although it seems that our tapping task resulted in fatigue on the target muscle, we did not detect any significant tremor on the target muscles using three-dimensional camera system. Most of the previous studies used accelerometer to the assessment tool for finger tremor (Huang et al. 2006; Takanokura and Sakamoto 2005), using three-dimensional infrared camera might not be sensitive enough to detect subtle changes of acceleration since all of our subjects were healthy young adults.

We also investigated whether there was any difference in the risk factors existed between the symptomatic and non-symptomatic group. We found participants with symptoms seems to spent longer time on computer work, have less time on recreation, have longer experience on computer use, and with higher psychological distress, which were similar to previous study (Jensen 2003; Lassen et al. 2005; Zakaria et al. 2002). However, only experience on computer use reached significant level. Both groups recruited from campus were about the same age. The symptomatic group had 7.6 years of experience on computer use while the non-symptomatic group only had 6.1 years of experience. It seems that the earlier they started to use computer, the easier they got the musculoskeletal symptoms of upper extremity.

About the relative ergonomic placement of the work station, we found most of the components that we investigated between groups did not reach the significant level. Kryger et al. (2003) conducted a large cohort study and suggested self-reported ergonomic workplace factors did not predict future forearm pain. However, it seems that there were more persons in the symptomatic group reported sitting with an abnormal chair height while they were working (symptomatic: non-symptomatic group = 10/15:7/15). Since we did not include a large number of subjects in this study, our results might only be a future reference.

For the muscle strength measurement, it was found that females in the symptomatic group had weaker strength of wrist extensors than the non-symptomatic group. Previous study has demonstrated that due to differences in anthropometry and hormones, female tends to have higher rate of CTD than male (Hales and Bernard 1996). It was also found that people in the symptomatic group significantly decreased the strength of FDS after the tapping task, but it was not found in the non-symptomatic group. Symptomatic subjects might be easier to reach to the fatigue stage or painful stage as compared to the non-symptomatic subjects. Therefore, they could not maintain their strength level.

We used two regression models to find out the weight of contribution on musculoskeletal symptoms from results of the physical examinations and questionnaire. The first regression equation revealed that median frequency of the FDS EMG at 25% of MVC was the most significant factor which can classify people into the symptomatic group compared to other results from physical examinations. This result could be explained by a cause-effect response. We asked the subjects to perform a low intensity, repetitive

mouse task whose intensity was lower but most close to 25% MVC. Similar study was conducted by Arihara and Sakamoto (1999). They found the shape of the EMG spectrum of the extensor digitorum communis gradually shift to a lower domain after fatigue. The second regression equation revealed that age, total time spent on computer and mouse position of computer were better factors to distinguish people with musculoskeletal symptoms compared to other measures such as BMI, chair height, or mental distress. Previous study suggested that intensive use of a mouse device, keyboard usage were the main risk factors for forearm pain (Kryger et al. 2003). However, they did not analyze as many factors as our study. Besides, since our study only included cross-sectional data, some of our finding might be cause of CTDUE (e.g. longer experience of computer use), some of them might be result after prolonged computer use (e.g. decrease of median frequency). If clinicians would like to set up a clinical prediction rule of CTDUE, further longitudinal study might be necessary. There are several factors which might be related to CTDUE. Whether there are other important factors such as high force demand or awkward posture which may contribute to CTDUE is still an area that needs to be searched.

Conclusion

The symptomatic computer users seem to have longer experience of computer use than the non-symptomatic users. After performing a tapping task, the symptomatic users decreased the median frequency of the FDS EMG and muscle strength to a larger extent than the non-symptomatic users. To classify people into the symptomatic group, median frequency of the FDS EMG at 25% MVC, age, total time spent on computer, and mouse position of computer are better predictors according to the results of this study. This study demonstrates a value for the development of a future clinical prediction rule of CTDUE.

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